

KilnEye: Large Scale Brick Kiln Detection from Satellite Imagery - supplementary

Anonymous Authors*
Anonymous Affiliation

1 Additional Results

We run the cross-validation experiments on the Google Maps Static dataset for Delhi and present the results in Table 1. We observe that YOLO-OBB variants are performing better than other models. We

Bounding box	Model	mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑	
		FCBK	Zigzag	FCBK	Zigzag
AA	Faster RCNN	0.24 ± 0.16	0.95 ± 0.01	0.15 ± 0.09	0.65 ± 0.01
	DETA	0.38 ± 0.17	0.95 ± 0.01	0.28 ± 0.12	0.65 ± 0.01
	yolov8x	0.26 ± 0.04	0.97 ± 0.01	0.19 ± 0.03	0.69 ± 0.01
	yolov7x	0.49 ± 0.15	0.95 ± 0.01	0.38 ± 0.11	0.65 ± 0.01
	yolov8x-worldv2	0.33 ± 0.15	0.97 ± 0.01	0.23 ± 0.10	0.70 ± 0.02
OBB	yolov8n	0.38 ± 0.12	0.96 ± 0.02	0.32 ± 0.11	0.81 ± 0.02
	yolov8m	0.36 ± 0.22	0.97 ± 0.01	0.28 ± 0.17	0.80 ± 0.04
	yolov8l	0.46 ± 0.11	0.97 ± 0.01	0.35 ± 0.09	0.82 ± 0.01
	yolov8x	0.51 ± 0.05	0.97 ± 0.01	0.41 ± 0.03	0.81 ± 0.01

Table 1: Performance of object detection models averaged over 4-fold cross-validation on the ‘Delhi’ Google Maps Static dataset. YOLO-OBB models outperform other models on this task.

run the cross-validation experiments on the Google Maps Static dataset for West Bengal and present the results in Table 2. We observe that YOLO-OBB variants are performing better than other models. We run the cross-validation experiments on the Sentinel Imagery for West Bengal and present the results in Table 3. We observe that YOLO-OBB variants are performing better than other models. We run the cross-validation experiments on the Sentinel Imagery for Lucknow and present the results in Table 4. We observe that YOLO-OBB variants are performing better than other models. We run the model trained on Google Maps Static imagery of Lucknow region on a distant region and present the results in Table 5. The models trained on Sentinel imagery do not yield encouraging results within the same domain, so we do not perform cross-domain evaluations with Sentinel imagery.

*Equal contribution

Bounding box	Model	mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑	
		FCBK	Zigzag	FCBK	Zigzag
AA	Faster RCNN	0.57 ± 0.11	0.66 ± 0.10	0.30 ± 0.07	0.35 ± 0.07
	DETA	0.66 ± 0.07	0.65 ± 0.10	0.36 ± 0.08	0.36 ± 0.09
	yolov8x	0.69 ± 0.06	0.69 ± 0.05	0.39 ± 0.04	0.39 ± 0.04
	yolov7x	0.36 ± 0.03	0.29 ± 0.11	0.18 ± 0.01	0.11 ± 0.05
	yolov8x-worldv2	0.68 ± 0.09	0.75 ± 0.09	0.42 ± 0.08	0.43 ± 0.07
OBB	yolov8n	0.70 ± 0.07	0.73 ± 0.03	0.51 ± 0.07	0.52 ± 0.04
	yolov8m	0.78 ± 0.04	0.77 ± 0.04	0.59 ± 0.05	0.56 ± 0.06
	yolov8l	0.78 ± 0.07	0.80 ± 0.04	0.59 ± 0.04	0.60 ± 0.05
	yolov8x	0.80 ± 0.02	0.80 ± 0.04	0.60 ± 0.04	0.61 ± 0.04

Table 2: Performance of object detection models averaged over 4-fold cross-validation on the ‘West Bengal’ Google Maps Static dataset. YOLO-OBB models outperform other models on this task.

Bounding box	Model	mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑	
		FCBK	Zigzag	FCBK	Zigzag
AA	Faster RCNN	0.36 ± 0.06	0.50 ± 0.03	0.13 ± 0.04	0.17 ± 0.00
	DETA	0.16 ± 0.06	0.24 ± 0.05	0.04 ± 0.01	0.07 ± 0.01
	yolov8x	0.30 ± 0.03	0.46 ± 0.12	0.09 ± 0.02	0.15 ± 0.04
	yolov7x	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
	yolov8x-worldv2	0.26 ± 0.05	0.43 ± 0.08	0.08 ± 0.02	0.14 ± 0.04
OBB	yolov8n	0.45 ± 0.02	0.65 ± 0.07	0.20 ± 0.02	0.31 ± 0.02
	yolov8m	0.51 ± 0.09	0.63 ± 0.08	0.24 ± 0.06	0.31 ± 0.02
	yolov8l	0.30 ± 0.04	0.54 ± 0.05	0.12 ± 0.02	0.24 ± 0.04
	yolov8x	0.34 ± 0.03	0.60 ± 0.04	0.14 ± 0.02	0.26 ± 0.03

Table 3: Performance of various models over 4-fold cross-validation on the West Bengal datasets with Sentinel imagery. YOLO-OBB models outperform other models on this task.

Bounding box	Model	mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑	
		FCBK	Zigzag	FCBK	Zigzag
AA	Faster RCNN	0.48 ± 0.06	0.66 ± 0.05	0.24 ± 0.03	0.34 ± 0.03
	DETA	0.23 ± 0.03	0.50 ± 0.05	0.08 ± 0.01	0.17 ± 0.02
	yolov8x	0.27 ± 0.02	0.59 ± 0.07	0.10 ± 0.03	0.24 ± 0.04
	yolov7x	0.13 ± 0.04	0.45 ± 0.07	0.05 ± 0.02	0.19 ± 0.03
	yolov8x-worldv2	0.33 ± 0.05	0.59 ± 0.02	0.13 ± 0.02	0.25 ± 0.02
OBB	yolov8n	0.51 ± 0.02	0.68 ± 0.06	0.30 ± 0.01	0.43 ± 0.04
	yolov8m	0.56 ± 0.03	0.71 ± 0.01	0.35 ± 0.03	0.46 ± 0.02
	yolov8l	0.34 ± 0.05	0.58 ± 0.06	0.17 ± 0.03	0.32 ± 0.04
	yolov8x	0.34 ± 0.05	0.60 ± 0.05	0.17 ± 0.04	0.34 ± 0.03

Table 4: Performance of various models over 4-fold cross-validation on the Lucknow datasets with Sentinel imagery.

Model	Delhi				West Bengal			
	mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑		mAP ₅₀ ↑		mAP ₅₀₋₉₅ ↑	
	FCBK	Zigzag	FCBK	Zigzag	FCBK	Zigzag	FCBK	Zigzag
Faster RCNN	0.28	0.80	0.19	0.46	0.25	0.47	0.12	0.26
DETA	0.30	0.85	0.22	0.54	0.55	0.49	0.28	0.25
yolov8x	0.35	0.80	0.26	0.51	0.42	0.54	0.22	0.27
yolov7x	0.15	0.50	0.11	0.29	0.47	0.40	0.24	0.18
yolov8x-worldv2	0.65	0.80	0.45	0.52	0.60	0.65	0.31	0.30
yolov8n	0.31	0.83	0.26	0.64	0.50	0.58	0.37	0.40
yolov8m	0.61	0.85	0.50	0.66	0.60	0.66	0.44	0.45
yolov8l-obbb	0.59	0.84	0.48	0.64	0.48	0.58	0.35	0.38
yolov8x-obbb	0.59	0.85	0.49	0.66	0.61	0.68	0.43	0.46

Table 5: Model performance on ‘Delhi’ and ‘West Bengal’ Google Maps Static datasets after training on ‘Lucknow’ Google Maps Static dataset.